

HSC Extension 2 Mathematics

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The Nature of Proof



Consider the following statement for $n \in \mathbb{Z}$.

If $n^2 + 4n$ is odd, then n is odd. Which of the following statements is the contrapositive of this statement for $n \in \mathbb{Z}$?

- A) If n is even, then $n^2 + 4n$ is even
- B) If $n^2 + 4n$ is even then n is even
- C) If n is odd then $n^2 + 4n$ is odd
- D) If $n^2 + 4n$ is odd then n is even



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The Nature of Proof



Given that a, b and c are positive real numbers, prove that

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$$(a + b + c)^2 \geq 3(ab + ac + bc)$$



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The Nature of Proof



Given that for $k > 0$, $\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0$ use Mathematical Induction to prove that for all integers $n \geq 2$,

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$$



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The Nature of Proof



Prove the following statement using a proof by contradiction.

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"For each irrational number s , the number $2s + 1$ is also irrational"



The Nature of Proof Solution

$$\begin{aligned}
 (a-b)^2 &\geq 0 \Rightarrow a^2 + b^2 \geq 2ab \\
 (b-c)^2 &\geq 0 \Rightarrow b^2 + c^2 \geq 2bc \\
 (a-c)^2 &\geq 0 \Rightarrow a^2 + c^2 \geq 2ac \quad ① \\
 2a^2 + 2b^2 + 2c^2 &\geq 2(ab + bc + ac) \\
 a^2 + b^2 + c^2 &\geq ab + bc + ac \quad ② \\
 (a+b+c)^2 - 2ab - 2bc - 2ac &\geq ab + bc + ac \\
 \therefore (a+b+c)^2 &\geq 3ab + 3bc + 3ac \quad ③
 \end{aligned}$$



The Nature of Proof Solution

A) If n is even, then $n^2 + 4n$ is even



The Nature of Proof Solution

Assume the statement is false
that for an irrational number s ,
there exists a number $2s+1$ which is
rational.
ie: $2s+1 = \frac{a}{b}$ ① where a and b are
integers with no common
factors.
 $\therefore 2s = \frac{a}{b} - 1$
 $= \frac{a-b}{b}$
 $2s = \frac{a-b}{b}$
 $s = \frac{a-b}{2b}$
 $\therefore s = \frac{c}{d}$ where c and d are
integers since a and b
are. ②
 $\therefore s$ is rational
CONTRADICTION ③
 $\therefore$ Original statement is true.



The Nature of Proof Solution

Show true for $n=2$

$$\begin{aligned}
 \frac{1}{1^2} + \frac{1}{2^2} &< 2 - \frac{1}{2} \\
 1 + \frac{1}{4} &< 1\frac{1}{2}
 \end{aligned}$$

so true for $n=2$

Assume true for $n=k$ where $k \geq 2$

$$\text{ie: } \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} < 2 - \frac{1}{k}$$

Now to show true for $n=k+1$

$$\begin{aligned}
 \text{Using assumption} \\
 \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \frac{1}{(k+1)^2} \quad ① \\
 < 2 - \frac{1}{k+1} + \frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1}
 \end{aligned}$$

$$< 2 - \frac{1}{k+1} \text{ given } \frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0$$

$\therefore$ True for $n=k+1$ ②

Since result is true for $n=2$, it
must also be true for $n=2+1=3$,
 $n=3+1=4$ and by induction for all
positive integral values of $n \geq 2$. ③



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The Nature of Proof

- (i) Given $f(t) = \sin(a + nt) \sin b - \sin a \sin(b - nt)$

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where a, b and n are constants with $a > 0, b > 0$ and $a + b < \pi$ and $n \neq 0$,

show that $f(t) = \sin(a + b) \sin nt$



- (ii) Find all values of t for which

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$$\frac{\sin(a+nt)}{\sin(b-nt)} = \frac{\sin a}{\sin b}$$



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The Nature of Proof

By expressing an odd integer as $2n - 1$ where n is a positive integer, prove that if we subtract one from the square of an odd number the result is a multiple of eight. [2]



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The Nature of Proof

Prove by contradiction that if p and q are positive integers, then

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$$\frac{p}{q} + \frac{q}{p} \geq 2.$$



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The Nature of Proof

It is proposed, for all integers x , that:

if $x^2 - 6x + 5$ is even, then x is odd.

- (i) Write down a contrapositive statement. [1]

- (ii) Use contraposition to prove the original conjecture. [2]



The Nature of Proof Solution

Let odd integer be $2n-1$, $n \in \mathbb{N}$
 Conjecture $(2n-1)^2 - 1$ is a multiple of 8
 $(2n-1)^2 - 1 = 4n^2 - 4n + 1 - 1$
 $= 4n^2 - 4n$
 $= 4n(n-1)$ ✓
 since $n \in \mathbb{Z}^+$, $n(n-1)$ is product of two consecutive integers
 one of which must be even and other odd hence product
 is EVEN i.e. $n(n-1) = 2m$ $m \in \mathbb{N}$
 $(2n-1)^2 - 1 = 4 \times 2m$
 $= 8m$
 hence a multiple of 8. ✓



The Nature of Proof Solution

$$\begin{aligned} \text{ci) } f(t) &= \sin(a+nt)\sin b - \sin a \sin(b-nt) \\ &= (\sin a \cos nt + \sin nt \cos a) \sin b - \sin a (\sin b \cos nt - \sin nt \cos b) \quad (1) \\ &= \cos a \sin b \sin nt + \sin a \cos b \sin nt \\ &= (\cos a \sin b + \sin a \cos b) \sin nt \\ &= \sin(a+b) \sin nt \quad (2) \end{aligned}$$

$$\begin{aligned} \text{cii) } \frac{\sin(a+nt)}{\sin(b-nt)} &= \frac{\sin a}{\sin b} \\ \therefore \sin b \sin(a+nt) &= \sin a \sin(b-nt) \\ \sin b \sin(a+nt) - \sin a \sin(b-nt) &= 0 \\ \Rightarrow \sin(a+b) \sin nt &= 0 \text{ using ci) (1)} \\ \text{as } \sin(a+b) &\neq 0, \\ \sin nt &= 0 \quad (2) \\ n+ &= k\pi \quad k=0, \pm 1, \pm 2, \dots \\ t &= \frac{k\pi}{n} \quad k=0, \pm 1, \pm 2, \pm 3, \dots \end{aligned}$$



The Nature of Proof Solution

(i) If $p \Rightarrow q$, Contrapositive is $\neg q \Rightarrow \neg p$
 If x is NOT odd then $x^2 - 6x + 5$ is NOT even ✓
 OR If x is EVEN then $x^2 - 6x + 5$ is ODD. ✓
 (ii) Let $x = 2n$, $n \in \mathbb{Z}$
 $x^2 - 6x + 5 = (2n)^2 - 6(2n) + 5$
 $= 4n^2 - 12n + 5$
 $= 4n(n-3) + 5$ ✓
 $= 2q + 5$ where $q \in \mathbb{Z}$ ✓
 hence $x^2 - 6x + 5$ is ODD ✓
 hence original conjecture proven by contraposition ✓



The Nature of Proof Solution

By way of contradiction, assume $\frac{p}{q} + \frac{q}{p} < 2$ ✓
 $(\times pq)$ since $p, q \in \mathbb{Z}^+$
 $p^2 + q^2 < 2pq$ ✓
 $p^2 - 2pq + q^2 < 0$ ✓
 $(p-q)^2 < 0$
 This is a contradiction since a square is by definition non-negative. ✓
 Hence $\frac{p}{q} + \frac{q}{p} \geq 2$ using proof by contradiction



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The Nature of Proof



Bernoulli's inequality famously states that,

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if $x \geq -1$ then $(1+x)^n \geq 1+nx$, for all positive integers n .

Prove this inequality using mathematical induction.



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The Nature of Proof



Given that $a, b \in \mathbb{N}$ and that a is a multiple of 6 and b is a multiple of 4, prove that $2a+3b$ is a multiple of 12.

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The Nature of Proof



Prove by contradiction that for rational non-zero a and b and irrational x , that $ax+b$ is irrational.

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The Nature of Proof



Prove that $x^{\frac{5}{2}} + 1 \geq x^2 + x^{\frac{1}{2}}$ for real $x \geq 0$.

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The Nature of Proof Solution

$$\exists n, k \in \mathbb{Z} \text{ such that } a=6n, b=4k.$$

$$\therefore 2a+3b = 12n+12k$$

$$= 12(n+k)$$

$$\therefore 2a+3b \text{ is a multiple of } 12.$$



The Nature of Proof Solution

Prove true for $n=1$

$$\text{LHS} = (1+x)^1 = 1+x$$

$$\text{RHS} = 1+x$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$

Assume true for $n=k$ $k \in \mathbb{Z}^+$

$$(1+x)^k \geq 1+kx$$

Prove true for $n=k+1$, that is

$$(1+x)^{k+1} \geq 1+(k+1)x$$

$$\text{LHS} = (1+x)^{k+1}$$

$$= (1+x)(1+x)^k$$

$$\geq (1+x)(1+kx)$$

$$= 1+x+kx+kx^2$$

$$= 1+(k+1)x+kx^2$$

$$\geq 1+(k+1)x \quad \text{since } x^2 \geq 0 \text{ and } k \in \mathbb{Z}^+$$

hence $\text{LHS} \geq \text{RHS}$
 hence if conjecture true for $n=k$ also true for $n=k+1$
 By principle of mathematical induction true for all positive integers n

(Note: The induction step is NOT TRUE for $x < -1$ and actually needs careful treatment for $-1 < x < 0$, considering signs but for majority purposes ignore this)



The Nature of Proof Solution

RTP: $x^{\frac{5}{2}} + 1 \geq x^2 + x^{\frac{1}{2}}$

$$\text{LHS} - \text{RHS} = x^{\frac{5}{2}} - x^2 - x^{\frac{1}{2}} + 1$$

$$= x^2(\sqrt{x}-1) - (\sqrt{x}-1)$$

$$= (\sqrt{x}-1)(x^2-1)$$

Natural domain is $x \geq 0$.

$\therefore$ for $0 \leq x < 1$, $(\sqrt{x}-1) < 0$ and $(x^2-1) < 0$

$\therefore (\sqrt{x}-1)(x^2-1) > 0$

for $x \geq 1$, $(\sqrt{x}-1) \geq 0$ and $(x^2-1) \geq 0$

$\therefore (\sqrt{x}-1)(x^2-1) \geq 0$

$\therefore \text{LHS} - \text{RHS} \geq 0$ for all real x in domain.

$\therefore x^{\frac{5}{2}} + 1 \geq x^2 + x^{\frac{1}{2}}$



The Nature of Proof Solution

Let $a = \frac{p}{q}$, $b = \frac{r}{t}$ where $p, q, r, t \neq 0 \in \mathbb{Z}$

Assume that $ax+b = \frac{u}{v}$ where $u, v \neq 0$

$u, v \in \mathbb{Z}$

$$\therefore \frac{p}{q}x + \frac{r}{t} = \frac{u}{v}$$

$$ptvx + rqv = uqt$$

$$x = \frac{uqt - rqv}{ptv}$$

Both the numerator and denominator are integers (non-zero integers are a closed set for multiplication and subtraction). $\therefore$ RHS is rational, but x is irrational, so there is a contradiction.

$\therefore ax+b$ is irrational.



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The Nature of Proof



Given that $u_{n+1} = 2u_n + 3$ and $u_1 = 1$, use mathematical induction to prove that $u_n = 2^{n+1} - 3$ for all positive integers n . 3



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The Nature of Proof



For all natural numbers n , if $(n-5)^2$ is an even integer, prove by contrapositive that n is odd. 2



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The Nature of Proof



Given that $\vec{a} = x\vec{i} + y\vec{j} + \frac{\sqrt{7}}{4}\vec{k}$, $\vec{b} = (x-2)\vec{i} + y\vec{j} + \frac{15}{4\sqrt{7}}\vec{k}$ and $\vec{a} \cdot \vec{b} = 0$, determine the value(s) of x and y . 3



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The Nature of Proof



Suppose that there exists some natural number $n = k$, such that $n^{n+1} > (n+1)^n$.

- (i) Use a proof by contradiction to show that this must be true also for $n = k + 1$, by substituting and then multiplying the inequalities. 2
- (ii) Find the smallest value of n for which $n^{n+1} > (n+1)^n$ holds, and then prove the inequality holds for all natural numbers greater than or equal to that value of n . 2



Contrapositive: RTP:

'If n is even then $(n-5)^2$ is odd.'

$\exists k \in \mathbb{Z} \mid n = 2k$

$$(2k-5)^2 = 4k^2 - 20k + 25$$

$$= 2(2k^2 - 10k + 12) + 1$$

$\therefore (2k-5)^2$ is odd.

$\therefore$ contrapositive statement is true

$\therefore$ if $(n-5)^2$ is even, n is odd.



$$u_{n+1} = 2u_n + 3, u_1 = 1.$$

Prove $u_n = 2^{n+1} - 3$.

For $n=1$, $u_1 = 1$

$$\text{and } u_1 = 2^{1+1} - 3$$

$$= 4 - 3$$

$$= 1$$

$\therefore$ true for $n=1$

Assume true for $n=k$,

$$\text{i.e. } u_k = 2^{k+1} - 3. \quad (*)$$

RTP true for $n=k+1$,

$$\text{i.e. } u_{k+1} = 2^{k+2} - 3$$

$$u_{k+1} = 2u_k + 3 \quad (\text{given})$$

$$= 2(2^{k+1} - 3) + 3 \quad (\text{using } *)$$

$$= 2^{k+2} - 6 + 3$$

$$= 2^{k+2} - 3$$

$\therefore$ true for $n=k+1$.

$\therefore$ true for all positive integers n by mathematical induction.



(i) $k^{k+1} > (k+1)^k$

Assume not true for $n=k+1$,

$$\text{i.e. } (k+1)^{k+2} \leq (k+2)^{k+1} \quad (1)$$

$$(k+1)^k < k^{k+1} \quad (2)$$

$$(1) \times (2): (k+1)^{2k+2} < k^{k+1} (k+2)^{k+1}$$

$$((k+1)^2)^{k+1} < (k^2 + 2k)^{k+1}$$

$$(k^2 + 2k + 1)^{k+1} < (k^2 + 2k)^{k+1}$$

Not true as $k^2 + 2k + 1 > k^2 + 2k$.

$\therefore$ true for $n=k+1$,

$$\text{i.e. } (k+1)^{k+2} > (k+2)^{k+1} \quad \text{if } k^{k+1} > (k+1)^k$$

(ii) For $n=1$, $1^2 < 2^1$

$n=2$, $2^3 < 3^2$

$n=3$, $3^4 > 4^3$

$\therefore$ true for $n=3$.

Assume true for $n=k$,

$$\text{i.e. } k^{k+1} > (k+1)^k$$

$\therefore$ true for $n=k+1$,

$$\text{i.e. } (k+1)^{k+2} > (k+2)^{k+1} \quad \text{from (i)}$$

$\therefore$ true for all $n \geq 3$ by mathematical induction.



$|a|=1$ by definition.

$$\therefore x^2 + y^2 + \frac{7}{16} = 1$$

$$x^2 + y^2 = \frac{9}{16} \quad (1)$$

$$\text{Also, } a \cdot b = 0$$

$$x^2 - 2x + y^2 + \frac{15}{16} = 0$$

$$(x-1)^2 - 1 + y^2 + \frac{15}{16} = 0$$

$$(x-1)^2 + y^2 = \frac{1}{16}$$



$$\therefore x = \frac{3}{4}, y = 0$$

$$\text{OR } (1) - (2): 2x - \frac{15}{16} = \frac{9}{16}$$

$$2x = \frac{24}{16} = \frac{3}{2}$$

$$\text{Sub into (1): } \left(\frac{3}{4}\right)^2 + y^2 = \frac{9}{16}$$

$$y^2 = 0, y = 0 \quad \therefore x = \frac{3}{4}, y = 0$$



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The Nature of Proof



Consider the statement

' $\exists x \in \mathbb{R}, \ln x = 1$ and $x > 2$.'

Which of the following is the negation of the statement?

- (A) $\exists x \in \mathbb{R}, \ln x \neq 1$ and $x \leq 2$
- (B) $\exists x \in \mathbb{R}, \ln x \neq 1$ or $x \leq 2$
- (C) $\forall x \in \mathbb{R}, \ln x \neq 1$ and $x \leq 2$
- (D) $\forall x \in \mathbb{R}, \ln x \neq 1$ or $x \leq 2$



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The Nature of Proof



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Disprove the statement:

'There exists $b \in \mathbb{N}$ such that $b^2 + 9b + 20$ is a prime number.'



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The Nature of Proof



' mnk ' represents a three-digit number, where m, n and k are the 1st, 2nd and 3rd digit respectively. If $m + n + k$ is divisible by three, show that mnk is divisible by three.



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The Nature of Proof



Prove by contradiction that there are no positive integers p and q , such that $4p^2 - q^2 = 25$.



The negation statement is:

"For all $b \in \mathbb{N}$, $b^2 + 9b + 20$ is not prime."

Now

$$b^2 + 9b + 20 = (b+4)(b+5)$$

But the product of two numbers is a composite number, that is, it's not a prime number.

$\therefore$ Since the negation statement is true then the original statement is not true $\therefore$ disproved.

Note: This question was not answered well.

Many students used counterexamples to disprove the statement.

You can't use counterexamples to disprove a statement that says 'there exists....'

To disprove this statement successfully with counterexamples, you will have to use ALL the numbers that make the statement false, and that is impossible to do unless you have finite numbers



$$\exists x \in \mathbb{R}, \ln x = 1 \text{ and } x > 2$$

Negation

$$\forall x \in \mathbb{R}, \ln x \neq 1 \text{ or } x \not> 2$$

D



Suppose there are positive integers p and q such that $4p^2 - q^2 = 25$

$$\therefore (2p-q)(2p+q) = 25$$

Since p and q are integers, $(2p-q)$ and $(2p+q)$ are also integers

Since the integer factors of 25 are 1, 5, and 25,

$(2p-q)$ and $(2p+q)$ are either 5 and 5, or 25 and 1.

If they are 5 and 5, then $q = 0$, which is a contradiction ($q > 0$)

$$\therefore 2p+q = 25 \text{ and } 2p-q = 1$$

$$\therefore 4p = 26$$

$$p = \frac{13}{2} \text{ which is a contradiction (p is an integer)}$$

$\therefore$ the supposition is false

$\therefore$ there are no positive integers p and q such that $4p^2 - q^2 = 25$



$$\text{Let } m+n+k = 3P \quad P \in \mathbb{Z}$$

(since $m+n+k$ is divisible by 3)

$$"mnk" = 100m + 10n + k$$

$$= 99m + 9n + m + n + k$$

$$= 99m + 9n + 3P$$

$$= 3(33m + 3n + P)$$

which is divisible by 3



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The Nature of Proof



For positive real number, x, y and z :

- (i) Show that $x + \frac{1}{x} \geq 2$. 1
- (ii) Hence show that $(x + y + z) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) \geq 9$. 2



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The Nature of Proof



Let n be a natural number. For each of the following statements, decide whether it is true or false. If true, give a proof. If false, give a counter example.

- (i) If n is a multiple of 4 then so is n^2 . 2
- (ii) If n^2 is a multiple of 4 then so is n . 2



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The Nature of Proof



For d , an integer when $d > 1$, show that:

- (i) $\frac{1}{d^2} < \frac{1}{d(d-1)}$. 1
- (ii) Noting that $\frac{1}{d^2-d} = \frac{1}{d-1} - \frac{1}{d}$, 2
- show that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2$ as $n \rightarrow \infty$.



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The Nature of Proof



We are given that $\frac{1}{a}$, $\frac{1}{b}$ and $\frac{1}{c}$ are three consecutive terms in a geometric series, where $a, b, c \in \mathbb{R}$. Show that $a^2 + c^2 \geq ab + bc$. 3



i) Statement is true

$$\text{Let } n = 4m, m \in \mathbb{Z}^+$$

Now

$$\begin{aligned} n^2 &= (4a)^2 \\ &= 16a^2 \\ &= 4 \times 4a^2 \end{aligned}$$

which is divisible by 4

$\therefore$ If n is a multiple of 4 then so is n^2

ii) Statement is false

Using a counter example

$$\text{Let } n^2 = 36 \text{ as}$$

36 is a multiple of 4.

but $\sqrt{36} = 6$ is not a

multiple of 4

$\therefore$ If n^2 is a multiple of 4 then so is n is a false statement.



i) Consider

$$(x-1)^2 \geq 0$$

$$x^2 - 2x + 1 \geq 0$$

$$x^2 + 1 \geq 2x$$

$$x + \frac{1}{x} \geq 2 \text{ since } x > 0$$

Inequality sign holds.

ii) Now

$$(x+y+z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) = 1 + \frac{x}{y} + \frac{x}{z} + \frac{y}{x} + 1 + \frac{y}{z} + \frac{z}{x} + \frac{z}{y} + 1$$

$$= 3 + \left(\frac{x}{y} + \frac{y}{x}\right) + \left(\frac{x}{z} + \frac{z}{x}\right) + \left(\frac{y}{z} + \frac{z}{y}\right)$$

$$\text{Let } x \rightarrow \frac{x}{y}, \text{ and } x \rightarrow \frac{x}{z}, x \rightarrow \frac{y}{z}$$

$$\begin{aligned} &= 3 + \left(x + \frac{1}{x}\right) + \left(x + \frac{1}{x}\right) + \left(x + \frac{1}{x}\right) \\ &\geq 3 + 2 + 2 + 2 \text{ from (i)} \\ &\geq 9 \end{aligned}$$



$$\text{GP: } \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \quad a, b, c \in \mathbb{R}$$

$$\text{Common ratio } r = \frac{\frac{1}{b}}{\frac{1}{a}} = \frac{\frac{1}{c}}{\frac{1}{b}}$$

$$\frac{1}{b} = \frac{1}{c}$$

$$\therefore ac = b^2 \text{ --- (1)}$$

Now, consider $(a-b)^2 = a^2 - 2ab + b^2 \geq 0$

$$a^2 + b^2 \geq 2ab \text{ --- (2)}$$

$$\text{Similarly, } b^2 + c^2 \geq 2bc \text{ --- (3)}$$

$$a^2 + c^2 \geq 2ac \text{ --- (4)}$$

$$\textcircled{2} + \textcircled{3} + \textcircled{4}: 2(a^2 + b^2 + c^2) \geq 2(ab + bc + ac)$$

$$a^2 + ac + c^2 \geq ab + bc + ac \text{ from (1)}$$

$$\therefore a^2 + c^2 \geq ab + bc, \text{ as required}$$



$$i) d > d-1$$

$$\therefore d^2 > d(d-1) \text{ (since } d > 1)$$

$$\therefore \frac{1}{d^2} < \frac{1}{d(d-1)}$$

$$ii) \text{ For } d=2: \frac{1}{2^2} < \frac{1}{2(2-1)} \text{ (from part i)}$$

$$\text{Similarly for } d=3: \frac{1}{3^2} < \frac{1}{3(3-1)}$$

$$d=n: \frac{1}{n^2} < \frac{1}{n(n-1)}$$

Summing:

$$\frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{(n-1)^2} + \frac{1}{n^2} < \frac{1}{2(2-1)} + \frac{1}{3(3-1)} + \dots + \frac{1}{(n-1)(n-2)} + \frac{1}{n(n-1)}$$

$$< \frac{1}{2} - \frac{1}{2} + \frac{1}{3} - \frac{1}{3} + \dots + \frac{1}{n-2} - \frac{1}{n-2} + \frac{1}{n-1} - \frac{1}{n-1} + \frac{1}{n} \text{ (from part ii)}$$

$$< 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n-2} - \frac{1}{n-2} + \frac{1}{n-1} - \frac{1}{n-1} + \frac{1}{n}$$

$$< 1 + \frac{1}{n}$$

$$\therefore \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 1 + \frac{1}{n}$$

$$< 2 + \frac{1}{n}$$

$$< 2 \text{ (as } \lim_{n \rightarrow \infty} \frac{1}{n} = 0)$$



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The Nature of Proof



- (i) Show that for $k \geq 2$, $\tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right) = \tan^{-1}\left(\frac{2}{k^2}\right)$ 2
- (ii) Hence, show that $\lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \tan^{-1}\left(\frac{2}{k^2}\right) \right] = \frac{3\pi}{4}$ 3



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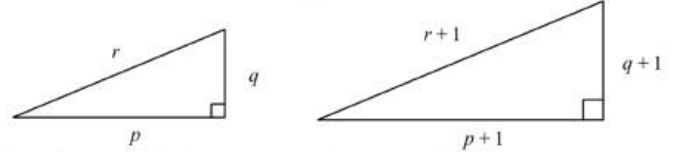
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The Nature of Proof



Consider the figure below, which shows two right-angled triangles. 3

NOT TO SCALE



Prove by contradiction that p , q and r cannot all be integers.



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The Nature of Proof



Consider the following statement, where x is an integer.

'If $x^2 - 6x + 5$ is even, then x is odd.'

- (i) Write down the contrapositive of the statement. 1
- (ii) Prove the statement by proving the contrapositive. 2



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The Nature of Proof



In a sequence of numbers, $T_1 = 7$ and $T_n = 2T_{n-1} - 1$ for $n \geq 2$. 3

Use mathematical induction to prove that $T_n = 3 \times 2^n + 1$ for $n \geq 1$.



The Nature of Proof Solution

Assume p, q, r are integers. Hence the Pythagorean Triads:

$$p^2 + q^2 = r^2 \quad \text{and} \quad (p+1)^2 + (q+1)^2 = (r+1)^2$$

$$\therefore p^2 + 2p + 1 + q^2 + 2q + 1 = r^2 + 2r + 1$$

$$r^2 + 2p + 2q + 2 = r^2 + 2r + 1$$

$$2(p+q) = 2r - 1$$

LHS even; RHS odd

Gives a contradiction, so the assumption that p, q , and r are integers must be false.



The Nature of Proof Solution

(i) RP: $\tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right) = \tan^{-1}\left(\frac{2}{k^2}\right)$ for $k \geq 2$

Consider $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 - \tan \alpha \tan \beta}$

$$= \frac{\frac{1}{k-1} - \frac{1}{k+1}}{1 + \left(\frac{1}{k-1}\right)\left(\frac{1}{k+1}\right)}$$

$$= \frac{\frac{k+1 - k-1}{(k-1)(k+1)}}{\frac{k^2 - 1 + 1}{(k-1)(k+1)}} = \frac{\frac{2}{(k-1)(k+1)}}{\frac{k^2}{(k-1)(k+1)}} = \frac{2}{k^2}$$

$\alpha = \tan^{-1}\left(\frac{1}{k-1}\right)$
 $\tan \alpha = \frac{1}{k-1}$
 $\beta = \tan^{-1}\left(\frac{1}{k+1}\right)$
 $\tan \beta = \frac{1}{k+1}$

$$\alpha - \beta = \tan^{-1}\left(\frac{2}{k^2}\right)$$

$$\therefore \tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right) = \tan^{-1}\left(\frac{2}{k^2}\right)$$

(ii) $\sum_{k=1}^n \tan^{-1}\left(\frac{2}{k^2}\right)$

$$= \tan^{-1} 2 + \sum_{k=2}^n \tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right)$$

$$= \tan^{-1} 2 + \underbrace{\tan^{-1}(1) - \tan^{-1}\left(\frac{1}{2}\right)}_{k=2} + \underbrace{\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{3}\right)}_{k=3} + \underbrace{\tan^{-1}\left(\frac{1}{3}\right) - \tan^{-1}\left(\frac{1}{4}\right)}_{k=4}$$

$$+ \underbrace{\tan^{-1}\left(\frac{1}{4}\right) - \tan^{-1}\left(\frac{1}{5}\right)}_{k=5} + \dots + \underbrace{\tan^{-1}\left(\frac{1}{n-2}\right) - \tan^{-1}\left(\frac{1}{n-1}\right)}_{k=n-1} + \underbrace{\tan^{-1}\left(\frac{1}{n-1}\right) - \tan^{-1}\left(\frac{1}{n}\right)}_{k=n}$$

$$= \tan^{-1} 2 + \cancel{\tan^{-1}(1)} + \cancel{\tan^{-1}\left(\frac{1}{2}\right)} - \cancel{\tan^{-1}\left(\frac{1}{2}\right)} + \cancel{\tan^{-1}\left(\frac{1}{3}\right)} - \cancel{\tan^{-1}\left(\frac{1}{3}\right)} + \dots + \cancel{\tan^{-1}\left(\frac{1}{n-1}\right)} - \cancel{\tan^{-1}\left(\frac{1}{n-1}\right)} + \tan^{-1}\left(\frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \tan^{-1}\left(\frac{2}{k^2}\right) \right] = \lim_{n \rightarrow \infty} \left(\frac{\pi}{4} + \tan^{-1} 2 + \tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right) \right)$$

$$= \frac{\pi}{4} + \tan^{-1} 2 + \lim_{n \rightarrow \infty} \left(\tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right) \right)$$

$$= \frac{\pi}{4} + \frac{\pi}{2} - 0 = \frac{3\pi}{4}$$

or $\dots \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}, x > 0$



The Nature of Proof Solution

RTP: $T_n = 3 \times 2^n + 1$

Step 1: Prove true for $n=1$.

$$T_1 = 7 \quad 3 \times 2^1 + 1 = 7 \quad \text{Hence true for } n=1.$$

Step 2: Assume: $T_k = 3 \times 2^k + 1$

$$\text{Prove: } T_{k+1} = 3 \times 2^{k+1} + 1$$

$$LHS = 2(T_k) - 1$$

$$= 2(3 \times 2^k + 1) - 1$$

$$= 3 \times 2^{k+1} + 2 - 1$$

$$= 3 \times 2^{k+1} + 1 = RHS$$

Thus proven by mathematical induction.



The Nature of Proof Solution

- (i) If x is even, then $x^2 - 6x + 5$ is odd
- (ii) Suppose that x is even, ie $x = 2k$ for integer k .

$$x^2 - 6x + 5 = (2k)^2 - 6(2k) + 5$$

$$= 4k^2 - 12k + 5$$

$$= 2(2k^2 - 6k + 2) + 1$$

$$= 2m + 1, \text{ where } m \text{ is an integer}$$

Hence, $x^2 - 6x + 5$ is odd and the statement is proven by proving the contrapositive.



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The Nature of Proof



Recall that $x + \frac{1}{x} \geq 2$ for any real number $x > 0$. **(DO NOT PROVE THIS)**

- (i) Prove that $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$
for all real numbers $a > 0, b > 0, c > 0$.

2

- (ii) Hence prove that $\left(\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b}\right) \geq \frac{3}{2}$
for all real numbers $a > 0, b > 0, c > 0$.

3



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The Nature of Proof



Use mathematical induction to prove that the number of diagonals of a convex polygon with n vertices is $\frac{1}{2}n(n-3)$ for $n \geq 4$. **3**



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The Nature of Proof



- (i) Prove that if x and y are positive real numbers, then $\frac{x+y}{2} \geq \sqrt{xy}$.

1

- (ii) The numbers $a_1, a_2, a_3, \dots, a_n$ are positive and real, and $a_1 a_2 a_3 \dots a_n = 1$.

2

Show that $(1+a_1)(1+a_2)(1+a_3) \dots (1+a_n) \geq 2^n$.



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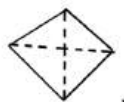
The Nature of Proof



Use mathematical induction to show that $\left(2 - \frac{1}{n}\right)^n > n$ for all integers $n \geq 2$. **3**



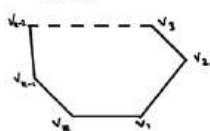
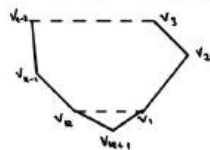
The Nature of Proof Solution

base case: $n=4$ 

Any convex quadrilateral will have 2 diagonals.

when $n=4$,

$$\frac{1}{2}(4)(4-3) = 2(1) = 2$$

 $\therefore$ true for $n=4$ Assume true for $n=k$,ie. any convex k -sided polygon will have $\frac{1}{2}k(k-3)$ diagonals.Prove true for $n=k+1$,ie. any convex $(k+1)$ -sided polygon will have $\frac{1}{2}(k+1)(k-2)$ diagonals.When one more side is added, a new vertex is formed. The $(k+1)$ th vertex can form a line with the k other vertices except for v_k and v_1 as this becomes the sides ie. $k-2$ new diagonals.The side v_k to v_1 will now become a diagonal as well.

$$\therefore \text{total new diagonals} = k-2+1 = k-1$$

$$\therefore \text{total diagonals} = \text{diagonal in } k\text{-sided polygon} + k-1$$



The Nature of Proof Solution

(i)

$$\begin{aligned} (a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) &= 1+\frac{a}{b}+\frac{a}{c}+1+\frac{b}{c}+\frac{c}{a}+1+\frac{c}{b} \\ &= 3+\left(\frac{a}{b}+\frac{b}{a}\right)+\left(\frac{a}{c}+\frac{c}{a}\right)+\left(\frac{b}{c}+\frac{c}{b}\right) \\ &\geq 3+2+2+2=9 \end{aligned}$$

 $\therefore LHS \geq RHS$ (ii) From (i), let $a \rightarrow a+b$ $b \rightarrow b+c$ $c \rightarrow a+c$

$$((a+b)+(b+c)+(a+c))\left(\frac{1}{a+b}+\frac{1}{b+c}+\frac{1}{a+c}\right) \geq 9$$

$$1+\frac{a+b}{b+c}+\frac{a+b}{a+c}+\frac{b+c}{a+b}+1+\frac{b+c}{a+c}+\frac{a+c}{a+b}+\frac{a+c}{b+c}+1 \geq 9$$

$$\frac{a+b+2c}{a+b}+\frac{2a+b+c}{b+c}+\frac{a+2b+c}{a+c} \geq 6$$

$$1+\frac{2c}{a+b}+1+\frac{2a}{b+c}+1+\frac{2b}{a+c} \geq 6$$

$$\rightarrow \frac{a}{b+c}+\frac{b}{a+c}+\frac{c}{a+b} \geq \frac{3}{2}$$



The Nature of Proof Solution

$$\left(2-\frac{1}{n}\right)^n > n, \quad n \geq 2$$

(A) Prove true for $n=2$:

$$\begin{aligned} LHS &= \left(2-\frac{1}{2}\right)^2 \\ &= \left(\frac{3}{2}\right)^2 \\ &= \frac{9}{4} \end{aligned}$$

$$\begin{aligned} RHS &= 2 \\ &< \frac{9}{4} \end{aligned}$$

 $\therefore$ true for $n=2$ (B) Assume true for some $n=k$, $k \geq 2$

$$\text{i.e. } \left(2-\frac{1}{k}\right)^k > k$$

Prove true for $n=k+1$:

$$\text{RTP: } \left(2-\frac{1}{k+1}\right)^{k+1} > k+1$$

$$\begin{aligned} LHS - RHS &= \left(2-\frac{1}{k+1}\right)^{k+1} - (k+1) \\ &> \left(2-\frac{1}{k}\right)^{k+1} - (k+1) \quad \text{as } k > 0 \\ &= \left(2-\frac{1}{k}\right)^k \left(2-\frac{1}{k}\right) - (k+1) \\ &> k\left(2-\frac{1}{k}\right) - (k+1) \quad \text{from (A)} \\ &= 2k-1-k-1 \\ &= k-2 \\ &\geq 0 \quad \text{as } k \geq 2 \end{aligned}$$

$$\therefore LHS - RHS > 0 \quad \text{and} \quad LHS > RHS$$

From (A) and (B) above, statement holds for all integers $n \geq 2$ by mathematical induction.

The Nature of Proof Solution

$$(i) \quad (\sqrt{x}-\sqrt{y})^2 \geq 0$$

$$\begin{aligned} \therefore x+y-2\sqrt{xy} &\geq 0 \\ x+y &\geq 2\sqrt{xy} \\ \frac{x+y}{2} &\geq \sqrt{xy} \end{aligned}$$

Note: Squaring both sides of an inequality was penalised if proper justification wasn't given.

$$\begin{aligned} (ii) \text{ Using (i): } 1+a_1 &\geq 2\sqrt{a_1} \\ 1+a_2 &\geq 2\sqrt{a_2} \\ 1+a_3 &\geq 2\sqrt{a_3} \\ &\vdots \\ 1+a_n &\geq 2\sqrt{a_n} \end{aligned}$$

(any correct application of AM-GM)

Multiplying:

$$\begin{aligned} (1+a_1)(1+a_2)(1+a_3)\dots(1+a_n) &\geq 2\sqrt{a_1} \times 2\sqrt{a_2} \times 2\sqrt{a_3} \times \dots \times 2\sqrt{a_n} \\ &\geq 2^n \sqrt{a_1 a_2 a_3 \dots a_n} \\ &\geq 2^n \quad \text{as } a_1 a_2 a_3 \dots a_n = 1 \end{aligned}$$



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The Nature of Proof



- (i) Let x and y be real numbers such that $x \geq 0$ and $y \geq 0$.

1

Prove that $\frac{x^2 + y^2}{2} \geq xy$.

- (ii) Suppose that a , b and c are real numbers, prove that $a^4 + b^4 + c^4 \geq a^2b^2 + a^2c^2 + b^2c^2$.

2

- (iii) Show that $a^2b^2 + a^2c^2 + b^2c^2 \geq a^2bc + b^2ac + c^2ab$.

2

- (iv) Deduce that if $a + b + c = d$, then $a^4 + b^4 + c^4 \geq abcd$.

1



HSC Extension 2 Mathematics

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The Nature of Proof



- (i) If $x \geq 0$, show that $\frac{x}{x^2 + 4} \leq \frac{1}{4}$.

2

- (ii) By integrating both sides of this inequality in part (i) with respect to x between the limits $x = 0$ and $x = \alpha$, show that

$$e^{\frac{1}{2}\alpha} \geq \frac{1}{4}\alpha^2 + 1 \text{ for } \alpha \geq 0.$$



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The Nature of Proof



It is known that $\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$. DO NOT PROVE THIS.

3

If x , y and z are the interior angles of a triangle, show that $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{\pi}$.



HSC Extension 2 Mathematics

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The Nature of Proof



Let $y = x^2 e^x$. Prove by mathematical induction that

3

$$\frac{d^n}{dx^n} x^2 e^x = (n(n-1) + 2nx + x^2) e^x \quad \forall n \in \mathbb{Z}^+.$$



The Nature of Proof Solution

$$(i) \quad (x-2)^2 \geq 0, \quad x \geq 0$$

$$x^2 - 4x + 4 \geq 0$$

$$x^2 + 4 \geq 4x$$

$$1 \geq \frac{4x}{x^2 + 4}$$

$$\frac{1}{4} \geq \frac{x}{x^2 + 4}$$

$$\therefore \frac{x}{x^2 + 4} \leq \frac{1}{4} \quad (\text{as required})$$

$$(ii) \quad \frac{x}{x^2 + 4} \leq \frac{1}{4} \quad [\text{from part (i)}]$$

$$\int_0^\alpha \frac{x}{x^2 + 4} dx \leq \int_0^\alpha \frac{1}{4} dx \quad \text{where } \alpha \geq 0$$

$$\frac{1}{2} \int_0^\alpha \frac{2x}{x^2 + 4} dx \leq \frac{1}{4} \int_0^\alpha 1 dx$$

$$\frac{1}{2} [\ln(x^2 + 4)]_0^\alpha \leq \frac{1}{4} [x]_0^\alpha$$

$$\ln(\alpha^2 + 4) - \ln 4 \leq \frac{1}{2} \alpha$$

$$\ln\left(\frac{\alpha^2 + 4}{4}\right) \leq \frac{1}{2} \alpha$$

$$\frac{\alpha^2 + 4}{4} \leq e^{\frac{1}{2}\alpha} \quad (\text{Since } f(x) = e^x \text{ is an increasing function})$$

$$\frac{\alpha^2}{4} + \frac{4}{4} \leq e^{\frac{1}{2}\alpha}$$

$$\therefore e^{\frac{1}{2}\alpha} \geq \frac{\alpha^2}{4} + 1 \quad (\text{as required})$$



The Nature of Proof Solution

$$(i) \quad \text{For all } x \geq 0, y \geq 0, \quad (x-y)^2 \geq 0$$

$$x^2 - 2xy + y^2 \geq 0$$

$$x^2 + y^2 \geq 2xy$$

$$\therefore \frac{x^2 + y^2}{2} \geq xy \quad (\text{as required})$$

ii) Substituting $x = a^2$ and $y = b^2$ into part (i),

$$\frac{(a^2)^2 + (b^2)^2}{2} \geq (a^2)(b^2)$$

$$\text{Similarly } \frac{b^4 + c^4}{2} \geq b^2 c^2 \quad \text{--- ②}$$

$$c^4 + a^4 \geq 2c^2 a^2 \quad \text{--- ③}$$

$$\frac{a^4 + b^4}{2} \geq a^2 b^2$$

$$\therefore a^4 + b^4 \geq 2a^2 b^2 \quad \text{--- ①}$$

$$\text{①} + \text{②} + \text{③} \quad 2a^4 + 2b^4 + 2c^4 \geq 2a^2 b^2 + 2b^2 c^2 + 2c^2 a^2$$

$$\therefore a^4 + b^4 + c^4 \geq a^2 b^2 + b^2 c^2 + c^2 a^2 \quad (\text{as req'd})$$

$$(iii) \quad \text{From part (i)} \quad x^2 + y^2 \geq 2xy \quad \text{Similarly } a^2(b^2 + c^2) \geq 2bca^2 \quad \text{--- ②}$$

$$\text{Let } x = a \text{ and } y = b, \quad a^2 + b^2 \geq 2ab$$

$$b^2(c^2 + a^2) \geq 2cab^2 \quad \text{--- ③}$$

$$c^2(a^2 + b^2) \geq 2abc^2 \quad \text{--- ①}$$

$$\text{①} + \text{②} + \text{③}$$

$$c^2(a^2 + b^2) + a^2(b^2 + c^2) + b^2(c^2 + a^2) \geq 2(abc^2 + bca^2 + cab^2)$$

$$c^2 a^2 + c^2 b^2 + a^2 b^2 + a^2 c^2 + b^2 c^2 + b^2 a^2 \geq 2(abc^2 + bca^2 + cab^2)$$

$$2a^2 b^2 + 2b^2 c^2 + 2c^2 a^2 \geq 2(abc^2 + bca^2 + cab^2)$$

$$\therefore a^2 b^2 + b^2 c^2 + c^2 a^2 \geq abc^2 + a^2 bc + b^2 ca \quad (\text{as required})$$

$$(iv) \quad \text{From part (ii)} \quad a^4 + b^4 + c^4 \geq a^2 b^2 + b^2 c^2 + c^2 a^2$$

$$\text{From part (iii)} \quad a^2 b^2 + b^2 c^2 + c^2 a^2 \geq c^2 ab + a^2 bc + b^2 ca$$

$$a^4 + b^4 + c^4 \geq a^2 b^2 + b^2 c^2 + c^2 a^2 \geq c^2 ab + a^2 bc + b^2 ca$$

$$\therefore a^4 + b^4 + c^4 \geq c^2 ab + a^2 bc + b^2 ca$$

$$\therefore a^4 + b^4 + c^4 \geq cab(c + a + b)$$

$$\text{Since } a + b + c = d$$

$$\therefore a^4 + b^4 + c^4 \geq cabd \quad (\text{as required})$$



The Nature of Proof Solution

$$\begin{aligned} \text{For } n=1: LHS &= \frac{d}{dx} [x^2 e^x] & RHS &= (1(1-1) + 2(1)x + x^2)e^x \\ &= x^2 e^x + 2xe^x & &= e^x (x^2 + 2x) \\ &= e^x (x^2 + 2x) & &= LHS \text{ hence true for } n=1. \end{aligned}$$

$$\text{Assume true for } n=k: \frac{d^k}{dx^k} [x^2 e^x] = (k(k-1) + 2kx + x^2)e^x \quad \forall k \in \mathbb{Z}^+$$

$$\text{R.T.P true for } n=k+1 \text{ i.e. r.t.p } \frac{d^{k+1}}{dx^{k+1}} [x^2 e^x] = ((k+1)(k) + 2(k+1)x + x^2)e^x \quad \forall k \in \mathbb{Z}^+$$

$$\begin{aligned} LHS &= \frac{d^{k+1}}{dx^{k+1}} [x^2 e^x] \\ &= \frac{d}{dx} \left[\frac{d^k}{dx^k} [x^2 e^x] \right] \\ &= \frac{d}{dx} [(k(k-1) + 2kx + x^2)e^x] \text{ from assumption} \\ &= (k^2 - k + 2kx + x^2)e^x + (2k + 2x)e^x \text{ by product rule} \\ &= (k^2 - k + 2kx + 2x + 2k + x^2)e^x \\ &= (k^2 + k + 2kx + 2x + x^2)e^x \\ &= ((k+1)(k) + 2(k+1)x + x^2)e^x \\ &= RHS \text{ as required} \end{aligned}$$

Hence true by the process of mathematical induction.



The Nature of Proof Solution

$$x + y + z = \pi \quad (\text{Interior angle sum of a triangle}) \quad [1] \quad \text{Given } \frac{x+y+z}{3} \geq \sqrt[3]{xyz} \quad [2]$$

Substituting into the given inequality [2]:

$$\frac{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}{3} \geq \sqrt[3]{\frac{1}{x} \cdot \frac{1}{y} \cdot \frac{1}{z}}$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{3}{\sqrt[3]{xyz}} \quad [3]$$

Consider the original inequality [2]:

$$\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$$

$$\frac{x+y+z}{9} \geq \frac{\sqrt[3]{xyz}}{3}$$

$$\frac{3}{\sqrt[3]{xyz}} \geq \frac{9}{x+y+z}$$

$$\text{Hence } \frac{3}{\sqrt[3]{xyz}} \geq \frac{9}{\pi} \quad (\text{from [1]}) \quad [4]$$

Now applying result [4] to [3]:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{\pi} \text{ as required.}$$



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The Nature of Proof



- (i) By considering the expansion of $k^3 - (k-1)^3$ show that:

$$\sum_{k=1}^n k^2 = \frac{n}{6}(n+1)(2n+1) \quad 2$$

(do **NOT** use mathematical induction)

- (ii) Prove that there are no finite sets of prime numbers where $2^2 + 3^2 + \dots + n^2$ is a multiple of at least one element of the set for all integers $n \geq 2$. 2



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The Nature of Proof



Show that:

3

$$|\cos(\alpha - \beta)| \leq 2$$

where $0 \leq \alpha \leq 2\pi$ and $0 \leq \beta \leq 2\pi$.



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The Nature of Proof



Suppose that $x + y = k$ for real numbers x, y and k .

- (i) Show that: 1

$$x + y \geq 2\sqrt{xy}$$

- (ii) If $x > 0$ and $y > 0$ prove that: 1

$$xy \leq \frac{k^2}{4}$$

- (iii) Hence, or otherwise, prove that: 3

$$\frac{1}{x^2} + \frac{1}{x} + \frac{1}{y^2} + \frac{1}{y} \geq \frac{4k+8}{k^2}$$



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The Nature of Proof



- (i) Show that $a^2 + 9b^2 \geq 6ab$, where a and b are real numbers. 1

- (ii) Hence, or otherwise, show that $a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac)$. 2

- (iii) Hence if $a > b > c > 0$, show that $a^2 + 5b^2 + 9c^2 > 9bc$. 2



The Nature of Proof Solution

$$\begin{aligned}
 |\cos(\alpha - \beta)| &\leq |\cos \alpha \cos \beta + \sin \alpha \sin \beta| \\
 &\leq |\cos \alpha \cos \beta| + |\sin \alpha \sin \beta| \quad (\text{triangle inequality}) \\
 &= |\cos \alpha| |\cos \beta| + |\sin \alpha| |\sin \beta| \\
 &\leq 2
 \end{aligned}$$

since $|\cos \alpha| \leq 1$, $|\cos \beta| \leq 1$, $|\sin \alpha| \leq 1$ and $|\sin \beta| \leq 1$.



The Nature of Proof Solution

$$\begin{aligned}
 k^3 - (k-1)^3 &= k^3 - (k^3 - 3k^2 + 3k - 1) \\
 &= 3k^2 - 3k + 1 \\
 \therefore 3k^2 &= k^3 - (k-1)^3 + 3k - 1 \\
 \therefore 3 \sum_{k=1}^n k^2 &= \sum_{k=1}^n k^3 - \sum_{k=1}^n (k-1)^3 + \sum_{k=1}^n (3k - 1) \\
 &\quad \text{telescoping} \quad \text{AP} \\
 3 \sum_{k=1}^n k^2 &= \cancel{1^3} - \cancel{0^3} + \cancel{2^3} - \cancel{1^3} + \cancel{3^3} - \cancel{2^3} + \dots + n^3 + (3n - 1) \\
 &\quad + \frac{1}{2} [2(2) + (n-1)3] \\
 &= n^3 + \frac{1}{2} [4 + 3n - 3] \\
 &= n^3 + \frac{1}{2} [3n + 1] \\
 &= \frac{1}{2} [2n^3 + 3n + 1] \\
 &= \frac{1}{2} [2n^3 + 2n + n + 1] \\
 &= \frac{1}{2} [2n(n+1) + 1(n+1)] \\
 3 \sum_{k=1}^n k^2 &= \frac{1}{2} (n+1)(2n+1) \\
 \sum_{k=1}^n k^2 &= \frac{1}{6} (n+1)(2n+1)
 \end{aligned}$$

(ii) Assume that such a set exists, where the primes are numbered:

$$p_1, p_2, p_3, \dots, p_n$$

$$\text{now, } 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} (n+1)(2n+1) - 1$$

(by rearranging part (i))

$$\text{let } S = p_1 p_2 \dots p_n$$

$$n = 65$$

$$\therefore 2^2 + 3^2 + \dots + n^2 = \frac{65}{6} (5+1)(25+1) - 1$$

$$= 5(5+1)(25+1) - 1$$

as $5(5+1)(25+1)$ is a multiple of all of the elements in the set

$5(5+1)(25+1) - 1$ is not a multiple of any element & there is always an n such that $2^2 + 3^2 + \dots + n^2$ is not a multiple of any element of the set



The Nature of Proof Solution

$$(i) \quad (a-3b)^2 \geq 0$$

$$a^2 - 6ab + 9b^2 \geq 0$$

$$a^2 + 9b^2 \geq 6ab$$

$$(ii) \quad a^2 + 9b^2 \geq 6ab \quad I$$

$$b^2 + 9c^2 \geq 6bc \quad II$$

$$a^2 + 9c^2 \geq 6ac \quad III$$

By Adding I, II and III

$$a^2 + 9b^2 + b^2 + 9c^2 + a^2 + 9c^2 \geq 6ab + 6bc + 6ac$$

$$2a^2 + 10b^2 + 18c^2 \geq 6(ab + bc + ac)$$

$$a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac)$$

$$(iii) \quad a > b > c > 0$$

$$ab > b^2 > bc > 0$$

$$ab > bc \quad I$$

$$a > b > c > 0$$

$$ac > bc > c^2 > 0$$

$$ac > bc \quad II$$

$$a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac) \quad \text{From Part ii)}$$

$$\geq 3(bc + bc + ac) \quad \text{From I}$$

$$\geq 3(bc + bc + bc) \quad \text{From II}$$

$$= 9bc$$

$$\therefore a^2 + 5b^2 + 9c^2 \geq 9bc$$



The Nature of Proof Solution

$$(i) \quad (\sqrt{x} - \sqrt{y})^2 \geq 0$$

$$x - 2\sqrt{xy} + y \geq 0$$

$$x + y \geq 2\sqrt{xy}$$

$$(ii) \quad \text{From (i): } x + y \geq 2\sqrt{xy}$$

$$2\sqrt{xy} \leq k$$

$$\sqrt{xy} \leq \frac{k}{2}$$

$$xy \leq \frac{k^2}{4}$$

$$(iii) \quad \text{From (i): } \frac{1}{x} + \frac{1}{y} \geq 2\sqrt{\frac{1}{xy}}$$

$$\text{From (ii): } \frac{1}{x} + \frac{1}{y} \geq 2\sqrt{\frac{4}{k^2}}$$

$$\frac{1}{x} + \frac{1}{y} \geq \frac{4}{k} \dots (1)$$

Similarly,

$$\frac{1}{x^2} + \frac{1}{y^2} \geq 2\sqrt{\frac{1}{x^2 y^2}}$$

$$\frac{1}{x^2} + \frac{1}{y^2} \geq \frac{2}{xy}$$

$$\frac{1}{x^2} + \frac{1}{y^2} \geq \frac{2}{\frac{k^2}{4}}$$

$$\frac{1}{x^2} + \frac{1}{y^2} \geq \frac{8}{k^2} \dots (2)$$

Adding (1) and (2):

$$\frac{1}{x^2} + \frac{1}{x} + \frac{1}{y^2} + \frac{1}{y} \geq \frac{4}{k} + \frac{8}{k^2}$$

$$\geq \frac{4k + 8}{k^2}$$

